C1.2 Cell respiration

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c1-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 412–424 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Glycolysis and anaerobic respiration

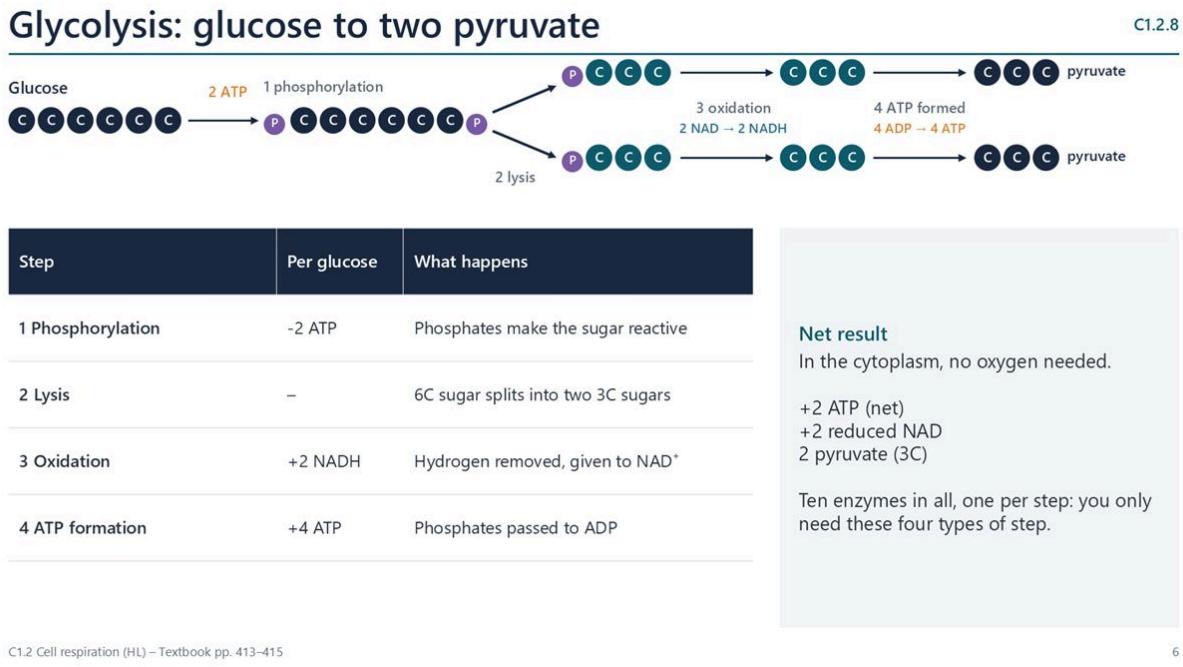
C1.2.7 Oxidation, reduction and NAD

pp. 412–413

Oxidation is loss of electrons; because a hydrogen atom carries an electron, removing hydrogen (**dehydrogenation**) oxidizes a substance. Respiration oxidizes glucose step by step. **NAD** accepts the hydrogen and is **reduced**, carrying it to the electron transport chain. Oxidation and reduction always occur together (redox).

C1.2.8 Glycolysis

pp. 413–415



In the cytoplasm, glucose is **phosphorylated** by 2 ATP, **split** into two 3C sugars, **oxidized** (2 NAD reduced) and converted to two **pyruvate**, making 4 ATP by substrate-level phosphorylation. Net yield: 2 ATP and 2 reduced NAD. Each step has its own enzyme.

C1.2.9 Lactate regenerates NAD

pp. 415–416

Without oxygen, reduced NAD cannot be oxidized by the electron transport chain. In human muscle, pyruvate accepts its hydrogen and becomes **lactate**, regenerating NAD+ so that glycolysis can continue. The net yield is only 2 ATP per glucose.

C1.2.10 Yeast, baking and brewing

pp. 416–417

Yeast uses the same glycolysis, but regenerates NAD by converting pyruvate to **ethanol and carbon dioxide**. In baking, carbon dioxide makes dough rise and the ethanol evaporates in the oven; in brewing, ethanol is the product wanted.

Aerobic respiration

C1.2.11 The link reaction

pp. 417–418

In the mitochondrial matrix, pyruvate is **decarboxylated** (CO_2 released) and **oxidized** (NAD reduced). The 2C acetyl group joins coenzyme A as acetyl-CoA. Fatty acids from lipids are also cut into acetyl groups that enter the same way.

C1.2.12 The Krebs cycle

pp. 418–419

The acetyl group joins **oxaloacetate** (4C) to form **citrate** (6C). Two decarboxylations and four oxidations (dehydrogenations) regenerate oxaloacetate. Each turn gives 2 CO_2, 3 reduced NAD, 1 reduced FAD and 1 ATP; the cycle turns twice per glucose.

C1.2.13 Reduced NAD and the electron transport chain

pp. 419–420

Reduced NAD from glycolysis, the link reaction and the Krebs cycle passes a pair of electrons to the first carrier of the chain in the inner membrane, becoming NAD+ again. This transfers its energy to the chain.

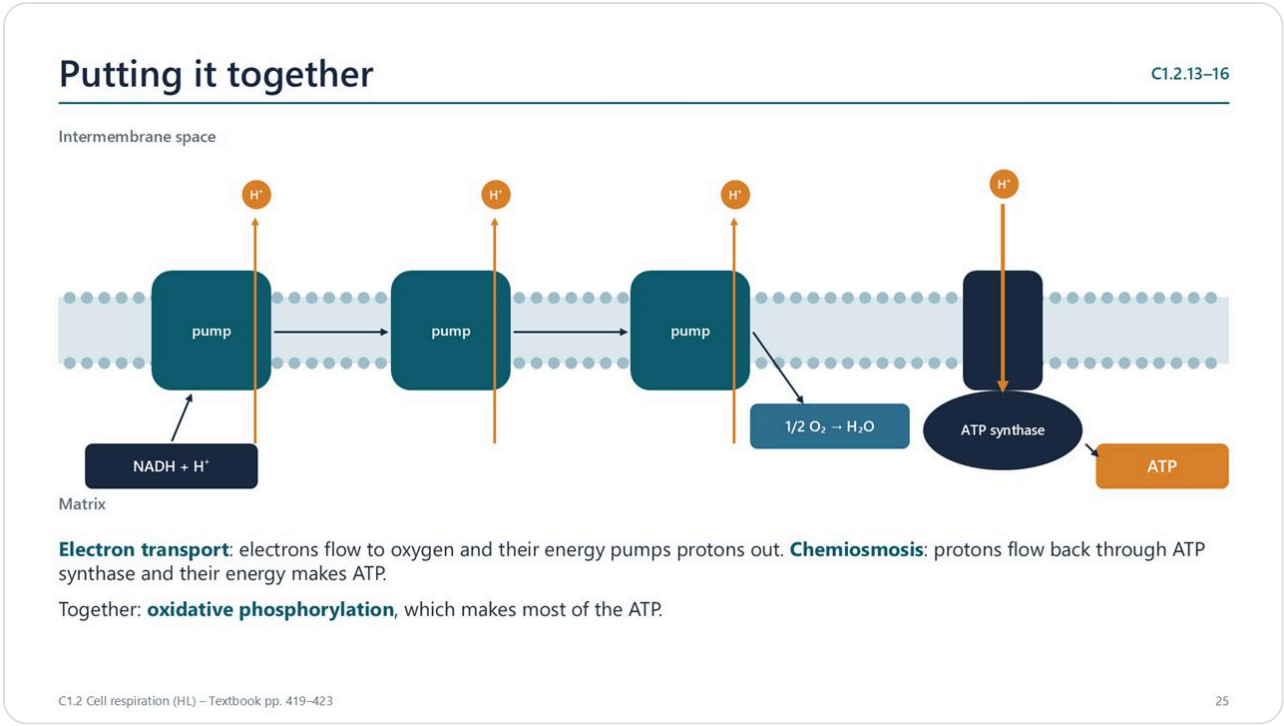
C1.2.14 The proton gradient

pp. 420–421

As electrons pass from carrier to carrier they release energy in small steps, which is used to pump **protons** from the matrix into the small intermembrane space. The inner membrane is impermeable to protons, so a steep gradient builds up.

C1.2.15 Chemiosmosis

pp. 421–422



Protons flow back into the matrix through **ATP synthase**. The flow turns part of the enzyme and provides energy to join ADP and phosphate: **oxidative phosphorylation**. This makes most of the ATP from aerobic respiration.

C1.2.16 Oxygen as the final acceptor

p. 422

At the end of the chain, **oxygen** accepts electrons and protons from the matrix, forming **metabolic water**. This keeps electrons flowing; without oxygen the chain stops and only glycolysis can produce ATP.

C1.2.17 Lipids and carbohydrates as fuels

pp. 423–424

Lipids give about twice as much energy per gram as carbohydrates, because they contain less oxygen and more hydrogen and carbon to oxidize. Fatty acids enter respiration as acetyl-CoA, not through glycolysis, so lipids can only be respired aerobically; carbohydrates can also be used anaerobically.

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